Exercises

Exercise 8.1 · Prime or not?. Are the following integers prime? 211; 221; 1,001; 257.

Exercise 8.2 · Factorisation and divisors. Factorise 720 into primes. How many positive divisors does it have? Are there many divisors of 720 that are perfect squares?

Beyond the syllabus · Least common multiple. The least positive common multiple of two nonzero integers, written lcm, is not part of the syllabus. It is used in the next exercise, which is not assessed.

Exercise 8.3 · GCD and LCM by factorisation. Factorise 1,260 and 1,050, then deduce their gcd and their least common multiple. Check that the product of the gcd and the lcm equals the product of the two numbers.

Exercise 8.4 · Fermat and remainders.

1. Determine the remainder of 2^{2026} in the division by 13.
2. Prove that for every integer n, $n^{13} - n$ is divisible by 13.

Exercise 8.5 · Divisible by 42. Prove that for every integer n, $n^7 - n$ is divisible by 42.

Exercise 8.6 · Square of a prime number. Let p be a prime number greater than or equal to 5. Prove that 24 divides $p^2 - 1$.

Exercise 8.7 · Mersenne numbers.

1. Using the factorisation of $z^m - 1$ (chapter 4), prove that if $n = ab$ with $a \geq 2$ and $b \geq 2$, then $2^a - 1$ divides $2^n - 1$.
2. Deduce that if $2^n - 1$ is prime, then n is prime.
3. Is the converse true? Compute $2^{11} - 1$ and check that 23 divides it.